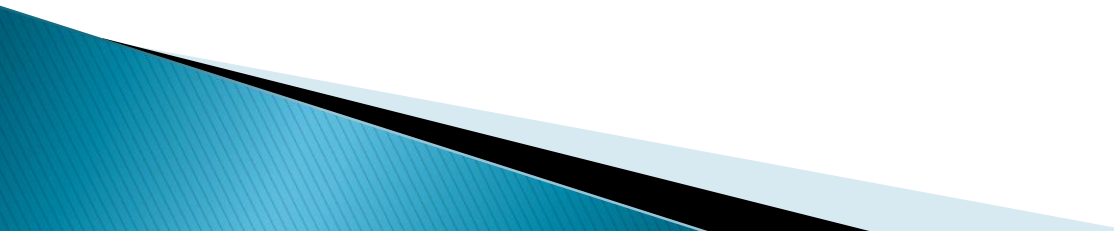


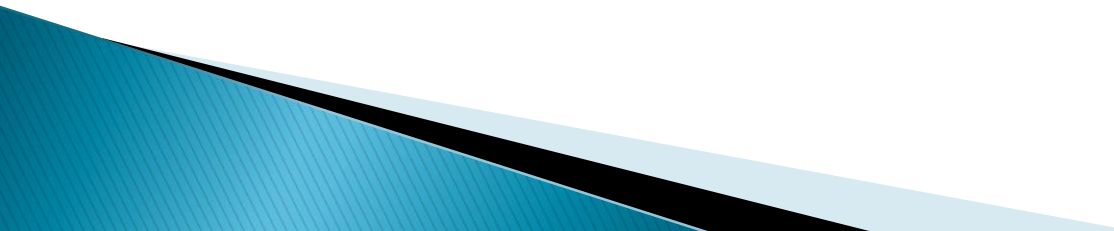
Side Effect Term Matching For Computational Adverse Drug Prediction

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Objective

- ▶ Develop predictive model for novel drugs
 - ▶ Form Data set
 - ▶ Edit distance
 - ▶ TRIE
 - ▶ KMP string matcher
 - ▶ Metamap sense resolution tool
 - ▶ Skip gram language model
- 

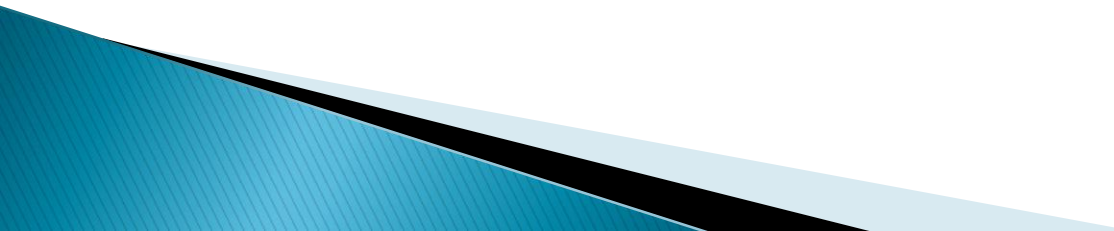
Data Set Formulation

- ▶ Minimal set of protein pharmacological profiling
 - ▶ Link Protein side effects with FDA approved Side Effect
 - ▶ Protein Dynamics and Docking Algorithm
 - ▶ Deep Neural Network
- 

Side Effect Matching

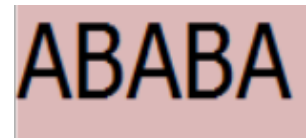
- ▶ Syntactically or semantic matching
- ▶ Possible to combine them

Edit Distance

- ▶ How similar two strings
 - ▶ Calculate the cost string transformation
 - ▶ Insertion, deletion and swapping operation
 - ▶ Dynamic programming technique
 - ▶ Complexity n^2
 - ▶ no involvement of semantic information
- 

KMP pattern matcher

- ▶ Pattern matching algorithm
- ▶ Efficiently find a pattern in a text document
- ▶ Use finite automaton
- ▶ Use longer string as a document
- ▶ Define a cost function
- ▶ Rank the text list
- ▶ Linear in time complexity
- ▶ No semantic information involve in whole operation



ABABA

Fig1: pattern

TRIE

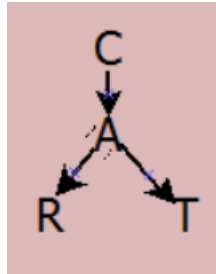


Fig3:

- Find LCA
- Calculate the cost based on the Depth
- Linear time
- Similar measurement like Edit distance

Why Syntactical Analysis

- ▶ Syntactical analysis easy
- ▶ Relatively small dataset
- ▶ Very fast and got quick result

Matching model

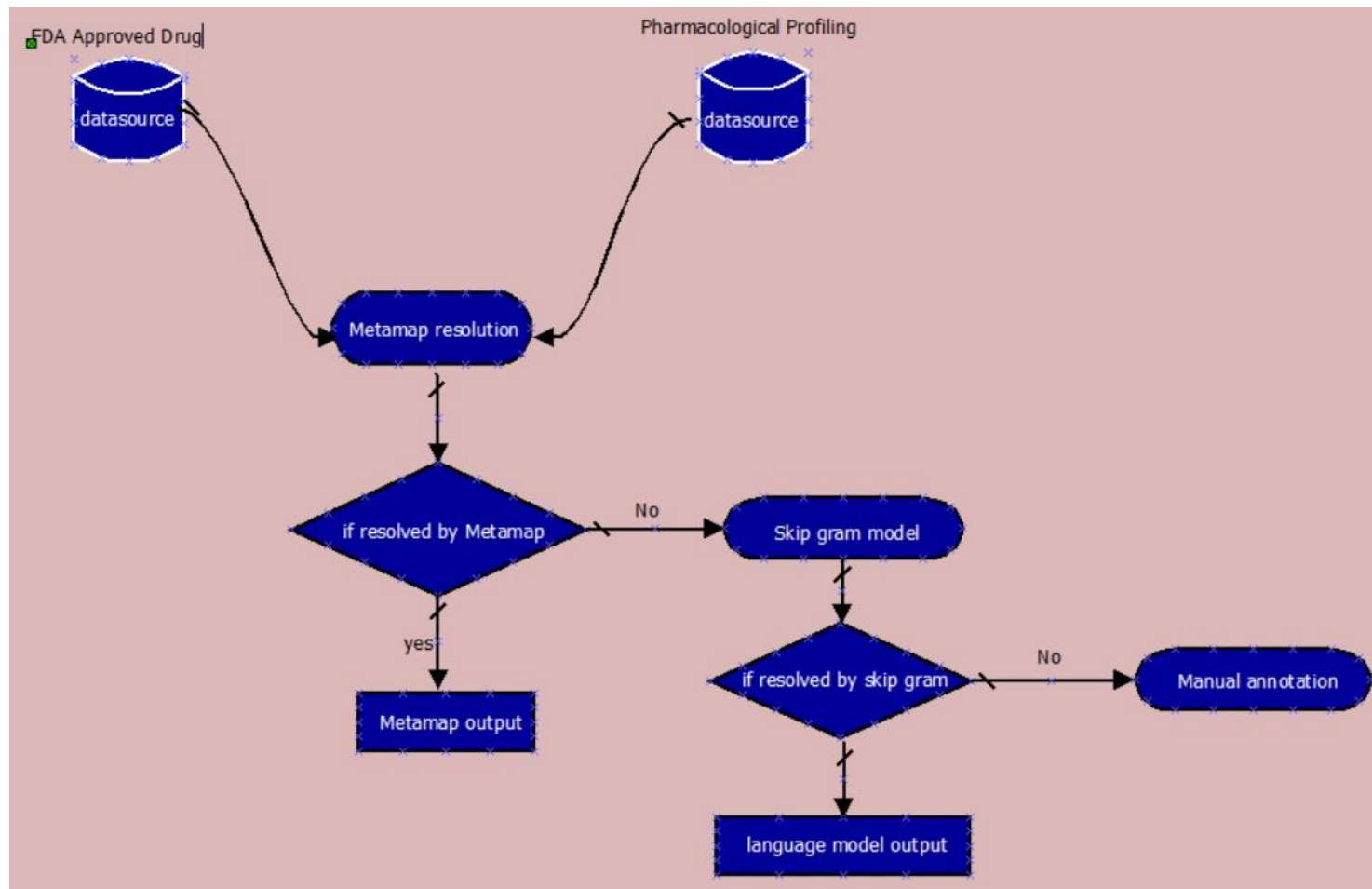


Fig3:

Metamap

- ▶ Metamap tool develop to access the UMLS knowledge base
- ▶ UMLS is the largest biomedical thesaurus
- ▶ UMLS has concepts dictionary, concept networks and special lexicons

Metamap Algorithm

- ▶ Parsing using xerox parsing
- ▶ Variant generation
- ▶ Candidate generation
- ▶ Candidate Evaluation

```
ocular {[adj], 0=""}  
  eye {[noun], 2="s"}  
    eyes {[noun], 3="si"}  
      optic {[adj], 4="ss"}  
        ophthalmic {[adj], 4="ss"}  
          ophthalmia {[noun], 7="ssd"}  
oculus {[noun], 3="d"}  
  oculi {[noun], 4="di"}
```

Fig4: Variants for Ocular

Metamap Algorithm

▶ Following four metrics

- Centrality
- Variation
- Coverage
- Cohesiveness

861 complications <1> (Complication)
861 complications <3> (Complications Specific to Antepartum or Postpartum)
777 Complicated
694 Ocular
638 Eye
638 Eye NEC
611 Ophthalmic
611 Optic (Optics)
588 Ophthalmia (Endophthalmitis)

Fig 2: Metathesaurus Candidate for Ocular Complications

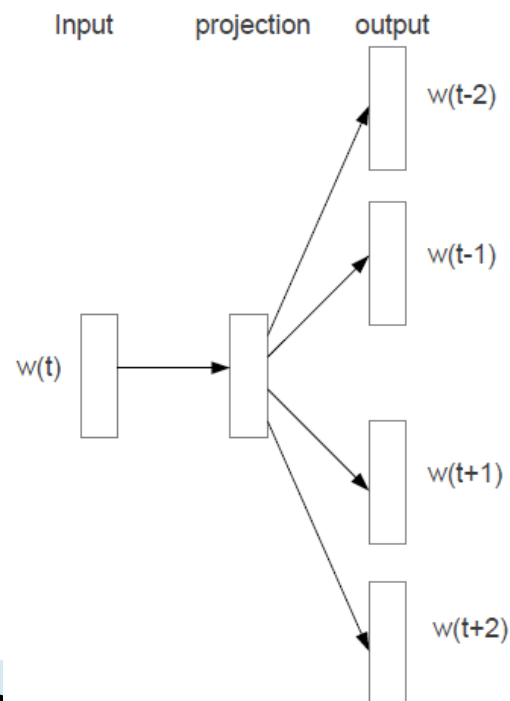
Metamap Result

Vitro Profiled side effect name	Matched FDA side effect name	Preferred Name	Concept cui
Potential cardiac valvulopathy	cardiac valve disease	heart valve disease	c0018824
cardiac hypertrophy	cardiac hypertrophy	cardiac hypertrophy	c1383860
in body weight	weight decreased	body weight	c0005910

Table I

Skip gram language model

- ▶ Distributed representation of words using vectors
- ▶ Avoid dense matrix multiplication
- ▶ This method encode the linguistic regularities



Language Model

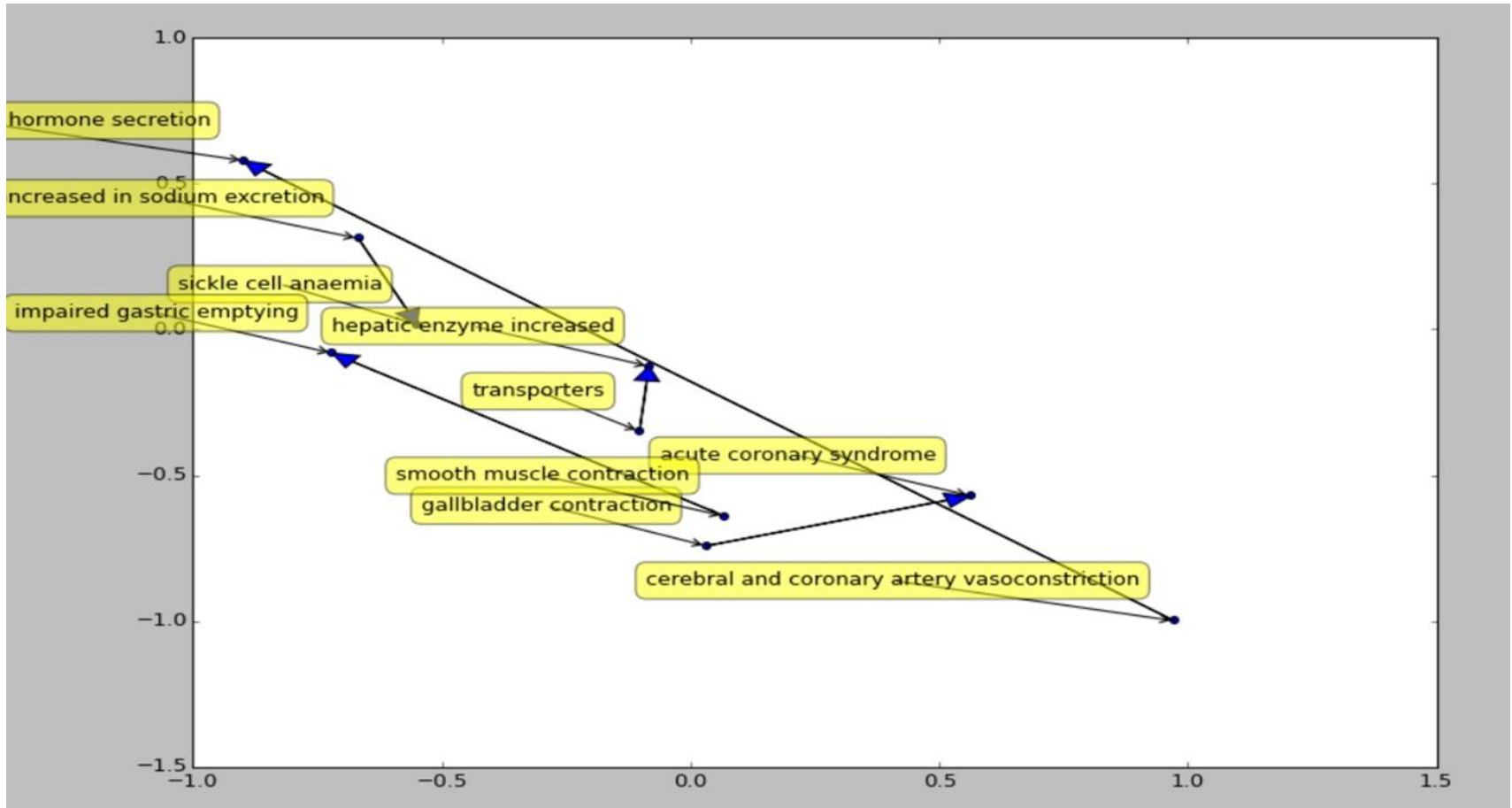
$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_{t+j} | w_t)$$

$$\text{logloss} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(p_{ij})$$

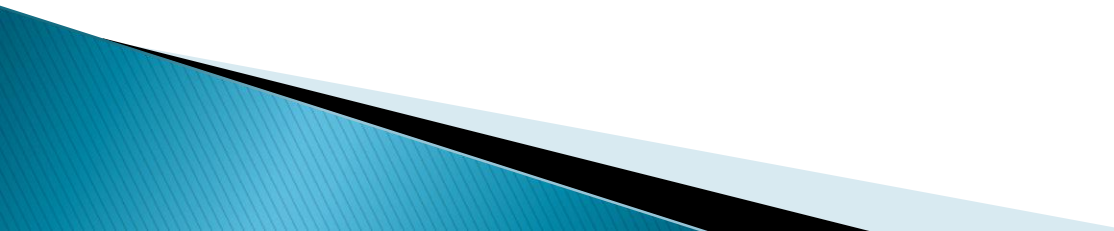
$$p(w_O | w_I) = \frac{\exp(v'_{w_O}{}^\top v_{w_I})}{\sum_{w=1}^W \exp(v'_w{}^\top v_{w_I})}$$

$$p(w | w_I) = \prod_{j=1}^{L(w)-1} \sigma \left(\mathbb{I}[n(w, j+1) = \text{ch}(n(w, j))] \cdot v'_{n(w, j)}{}^\top v_{w_I} \right)$$

Plot of word vectors



Conclusion

- ▶ Metamap resolved almost 75 percent of matching
 - ▶ Language model rest of the 20 percent out of 25
 - ▶ We need to do sense disambiguation
 - ▶ Complete the rest of the project
- 

Acknowledgement

Center for Clinical and Translational Sciences

NIH NCATS grant KL2TR000116

Markey Cancer Center, CRI SRF Center for Computational
Sciences (CCS)

Division of Biomedical Informatics, COM

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- ▶ Anthony Rios, Department of Computer Science, University of Kentucky

<https://github.com/AnthonyMRios/pymetamap>

